End-to-End SLA Estimation from Limited Provider Disclosures

Methods, Experimental Record, and Limits of Reconstruction

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Executive synthesis

In composed cloud-edge services, individual providers may advertise limited workload and SLA evidence while keeping their stochastic execution models private. The methodological challenge is to combine these separate disclosures into a prediction of end-to-end SLA satisfaction over a time horizon, without assuming access to hidden process parameters. We report the development, implementation, and evaluation of a family of methods indexed by provider information representation I and integration procedure M, together with a simulator and a complete account of observed limitations.

We define the evaluated quantity as $\sigma_G(A, H; \rho) = \Pr[c_G(A, H) \geq \rho]$, where $c_G(A, H)$ is the fraction of **decided** composed requests that meet a specified latency-cost-quality region by time H. A missed latency deadline is a decided failure; requests not yet decided are excluded; zero decided requests yield compliance one. This quantity measures the probability that the composed service meets its SLA compliance target over the observation horizon H.

Phase I (controlled pilot) tested the original three-provider parallel graph. The original controlled pilot was promising. Against a fresh white-box reference ($N_{\mathrm{WB}} = 200$: 200 new simulation runs using the true underlying model), across 555 related measurements (15 SLA conditions, each evaluated at 37 time horizons using shared simulation runs), the mean absolute errors (MAE) of a single reconstructed model (M1), a 27-member equal-weight ensemble (M2), and a Gibbs-weighted pilot M3 with 14 propagated joint models were respectively **0.18517, 0.06815 and 0.03037**. However, a retrospective cost-matched support audit found a Top1 estimator with MAE **0.02878**, slightly below the weighted Top14 estimator in that fixed pilot. The pilot therefore did not establish that multi-model averaging was necessary.

Phase II (expanded battery) evaluated generalization across multiple provider configurations and composition graphs. Sixteen evaluation cases were planned (four provider configurations combined with four composition graphs). The four cases involving configuration P2 did not meet the predefined criteria for informative SLA test conditions, so they were excluded without replacement. On the remaining 12 cells, with independent $N_{\mathrm{WB}} = 1000$ final white-box trajectories per cell and 6,660 related measurements (555 per case) per fully applicable method, the executed Phase-5 V3b algorithms produced MAE **0.15702** (M1), **0.13395** (M2), and **0.12189** (M3 FULL343). M3 retained the best aggregate ranking, but hard conditions still had substantial errors. Its algorithm and candidate support also changed from the pilot and from the initially registered Phase-5 method. Accordingly these numbers cannot be treated as direct replications of the original 14-member method.

Phase III (retrospective divergence tracing) investigates that gap, including an original-style M3 comparison, without creating a new independent holdout. Reconstructing 48 public-I1-scored candidates/provider and propagating Top14 at B=1400 on the same 12 cells gave **MAE 0.15416** (separately simulated Top1, **0.16228**), versus **0.12189** for executed V3b FULL343. Although Top14 retained at least **99.64%** of inferred joint-model weight in P1, P3 and P4, FULL343 outperformed it in **11 of 12** cells. These post-result comparisons do not isolate member count or constitute a fresh holdout.

Subsequent development tests revealed further limitations. On P4/G_SEQPAR, stronger reconstruction from I1 alone performed comparably to, and on average slightly better than, several richer I2 information-and-loss pipelines under matched search and graph-propagation designs. This does **not** isolate the causal utility of information quantity because fitting objectives also changed. In a separate P2/G_PAR experiment using a deterministically composed public graph region, M1, M2 and M3 achieved MAE **0.13200, 0.14058 and 0.12787**, respectively, under **unmatched** graph budgets. A privileged retrospective audit then found that the true generating provider parameters fitted the public I1 evidence better than the selected reconstructed candidates despite lying inside the search domain. This is direct evidence of a search/reconstruction deficit, not a proof of structural non-identifiability.

The overall conclusion is conditional: end-to-end SLA estimation from limited disclosure is demonstrably feasible **in some controlled cases**, but the developed finite-search integration methods have not delivered reliable prediction across the studied variations. The technical report retains favorable, negative, and inconclusive evidence and distinguishes

numerical Monte Carlo precision from uncertainty about the hidden system. In practical terms, this report provides the empirical foundation for a follow-up academic paper on how to design, validate, and report credible evidence for distributed management and optimization methods in hierarchical computing-continuum systems. The goal is to identify what must be controlled or tested before a promising pilot can support broader claims of robustness.

1. Research problem, scope, and prior context

1.1 Composable offerings and information boundaries

The computing continuum spans resources and services controlled by different administrative entities. An integrator may know its own workflow or service-composition graph while observing only restricted public claims about the providers on which it depends. Physical execution is shaped by stochastic service times, queues, branch synchronization, workload and cost accumulation. Predicting the performance of composed black-box services using only their providers' advertised performance measures is not equivalent to observing the resulting end-to-end behavior.

The Composable Autonomic Offering (CAO) abstraction represents an offering as

$$o = (G, s, \sigma, r), \tag{1}$$

where G is the composition graph, s describes service objectives, σ represents dependability or survival information, and r is the realization envelope [12]. The CAO abstraction motivates predicting a useful dependability quantity, but does not, by itself, prescribe which information must be disclosed or which numerical method can produce a dependable composition estimate.

We separate these choices explicitly:

$$\tau = (I, M), \tag{2}$$

with **information interface** I and **integration method** M. The interface I specifies what each provider discloses. Given these separate disclosures and the known composition graph G, the integrator applies M to estimate the composed service's end-to-end SLA probability, $\widehat{\sigma}_G(A, H; \rho)$. Depending on the method, this may involve reconstructing provider models and propagating their predictions through G. Distinct M may use the same I; conversely an information-rich interface can require a different reconstruction objective, making the evaluated unit an **information–method pair**.

The question is not whether analytical or stochastic QoS composition exists. It does [3,8,17–19]. Rather, the specific problem is **inverse, public-evidence-constrained** composition: can an integrator combine the probability surfaces advertised by individual black-box providers to estimate the composed service's finite-horizon, end-to-end SLA probability, without access to their execution traces or true parameters?

1.2 Research questions and evidence categories

The program addresses four related questions:

RQ1, feasibility and limits. When can finite public provider disclosures support useful graph-level SLA-satisfaction prediction, and in which worlds, topologies, regimes, or query regions does prediction fail?

RQ2, methods under a fixed interface. How do analytical propagation, a single provider reconstruction, equal-weight reconstruction portfolios and evidence-weighted finite supports compare? Which observed differences are due to candidate search, support membership, weights or numerical allocation, and which cannot be causally separated?

RQ3, disclosure. Does enriching the public interface with marginal or paired temporal compliance information improve the resulting *information–loss–integration pipeline* under matched reconstruction resources?

RQ4, provenance and decision meaning. How reproducible are the finite results, and what do numerical Monte Carlo intervals, model spread, reference error and support-truncation guarantees actually mean for an SLA threshold?

We organize the evidence into **three overarching phases: Phase I (controlled pilot), Phase II (expanded battery), and Phase III (retrospective divergence tracing)**. Phase I covers the frozen same-setting pilot; Phase II the registered expanded battery, documented pre-reference method revisions and held-out final evaluation; Phase III the retrospective divergence audit and related post-result exploratory analyses. These remain different evidence categories within the three phases. In Phase II, the final graph predictions were recorded before the final white-box results were examined, although the methods had already been revised during development. Phase III examines earlier results retrospectively and does not provide a new independent validation of those methods.

1.3 Scientific boundaries

This document does not establish a structural identifiability theorem. Multiple candidate models with different parameters and similar *finite empirical* I1 errors may reflect a non-injective information map, measurement noise, search limitations, model family restrictions, or some combination. The present observations cannot distinguish these causes in full generality.

2. Formal SLA objective and benchmark semantics

2.1 Request-level admissibility

For a composed request, denote latency, accumulated cost and quality by (L_G, C_G, Q_G). A rectangular admissibility region is

$$A = \{(L, C, Q) : L \leq \ell,\ C \leq c,\ Q \geq q\}.$$

A request is **compliant** when it is decided and its final outcome lies in A. The simulator records the top-level emission and, if available before simulation stop, the completion time. An on-time request is decided at completion. An uncompleted or late request is decided as a latency failure when its deadline arrives. A request whose completion and deadline are still in the future is unresolved at the current evaluation horizon. The SLA-scoring rule does not equate "unresolved" with "failed".

Let $\mathcal{D}_G(H)$ be the set of top-level requests *decided* by horizon H, and $\mathcal{C}_G(A, H) \subseteq \mathcal{D}_G(H)$ the compliant subset. Define, with the explicitly frozen zero-decision convention,

$$c_G(A, H) = \begin{cases} 1, & |\mathcal{D}_G(H)| = 0, \\ |\mathcal{C}_G(A, H)|/|\mathcal{D}_G(H)|, & |\mathcal{D}_G(H)| > 0. \end{cases} \tag{3}$$

For required compliance fraction $\rho \in [0, 1]$, the true graph-level estimand is

$$\boxed{\sigma_G(A, H; \rho) = \Pr\{c_G(A, H) \geq \rho\}.} \tag{4}$$

The probability is taken over independent stochastic trajectories of the specified provider world, workload and execution graph. The horizon H determines which decided requests are included in the compliance fraction. As more requests are decided, this fraction may

rise or fall, so $\sigma_G(A, H; \rho)$ is **not guaranteed to decrease with** H. We therefore use the precise term "horizon-dependent SLA-satisfaction probability" throughout the report.

The numerical methods compute $\widehat{\sigma}_G^{(I,M)}(A, H; \rho)$, and the hidden world is only evaluated through a separate finite white-box reference $\widehat{\sigma}_G^{\mathrm{WB}}$ under experimental conditions that permit it.

2.2 Prediction target and evaluation reference

What do we estimate? For a known composition graph G and an SLA query $(A, H; \rho)$, Equation (4) defines $\sigma_G(A, H; \rho)$, the probability that the composed service meets the required compliance fraction. The integrator does not know the providers' hidden stochastic execution models. It has only the information each provider discloses through interface I. Integration method M uses those disclosures and the known graph to produce the end-to-end prediction $\widehat{\sigma}_G^{(I,M)}(A, H; \rho)$. **The scientific task is to assess the accuracy and limits of this probability prediction.**

How do we evaluate it? In the simulator, the true provider models can be used to generate an independent finite white-box reference $\widehat{\sigma}_G^{\mathrm{WB}}(A, H; \rho)$. We compare each method's prediction with this reference across the tested SLA queries, horizons, provider worlds and graph topologies. The hidden provider models used for final reference generation are not supplied to the methods being evaluated. Because the reference is computed from a finite number of stochastic runs, it is itself an estimate of σ_G, not an exact value. Numerical simulation error and reconstruction error must therefore be distinguished (Chapter 10 and Appendix A).

3. Provider disclosures

3.1 Public interface I1

Under the original public interface I1, each provider advertises a finite family of rectangular admissibility regions indexed by a target content level ρ_r:

$$A_i(\rho_r) = \{L_i \le \ell_i(\rho_r),\, C_i \le c_i(\rho_r),\, Q_i \ge q_i\}. \tag{5}$$

Here L_i, C_i and Q_i are the provider's latency, cost and quality outcomes. The functions $\ell_i(\rho_r)$ and $c_i(\rho_r)$ specify **numerical latency and cost limits** for each advertised index ρ_r. They are finite lookup mappings, not time-varying performance functions. In this simulated experiment, the framework prepares a separate I1 card on behalf of each provider and makes it available to the integrator. To construct its advertised regions, the provider-side procedure fits a joint model of private latency and cost observations (a Gaussian mixture in log latency and log cost). It draws a fixed synthetic sample from that model and, for each ρ_r, chooses a minimum-area rectangle containing at least that fraction of sampled outcomes, while retaining the preceding region. Consequently the regions are **nested**: when $\rho_r < \rho_r'$, $A_i(\rho_r) \subseteq A_i(\rho_r')$, so neither the latency nor the cost limit decreases. In the studied provider worlds, the quality outcome is constant at $Q_i = 0.5$, with advertised limit $q_i = 0.5$. Thus quality is not a varied experimental dimension, and the regions differ only in their latency and cost limits.

The index ρ_r thus specifies the **target content of the fitted latency-cost model**, not a guaranteed fraction of compliant requests during an execution horizon. The latter is the separate SLA-compliance threshold ρ_q. Once the regions are fixed, each provider advertises an empirical probability surface

$$\widehat{\sigma}_i^{I1}(A_i(\rho_r), H; \rho_q) \approx \Pr\{c_i(A_i(\rho_r), H) \ge \rho_q\}. \tag{6}$$

The historical pilot used $\rho_r, \rho_q \in \{0.95, 0.975, 0.98333, 0.99, 0.995\}$ and a 5-second horizon grid up to 240 seconds, with a workload period of 0.2 seconds. Provider-card acquisition used two

disjoint banks of 100 complete simulation trajectories each. The first bank (seeds 6000–6099) was used only to fit the provider's latency-cost model and construct the regions. The second bank (seeds 6100–6199) evaluated the already fixed regions. For each $(A_i(\rho_r), H, \rho_q)$, each trajectory in the second bank contributed one binary outcome: whether its decided-request compliance fraction reached ρ_q by horizon H. Thus $\widehat{\sigma}_i^{I1} = n_{\text{pass}}/100$ is the fraction of successful trajectories, **not** an average of raw traces, the mean of per-trajectory compliance fractions, or the fraction of compliant requests pooled across trajectories. The same trajectories are reused across regions, horizons and compliance thresholds, so the points on the probability surface are correlated. Each simulation run also supplies provider-local ledgers for all three providers.

The public I1 card includes pointwise 95% Wilson binomial confidence intervals for each estimated SLA-satisfaction probability (the chance that a new independent provider trajectory achieves at least the required fraction ρ_q of compliant decided requests for region $A_i(\rho_r)$ by horizon H). These intervals are disclosed as part of I1, but none of the evaluated methods M0, M1, M2 or M3 uses them. The card does **not** disclose request timestamps, hidden queue/service trajectories or provider parameters.

The provider-side procedure uses private simulation evidence to construct the finite I1 card. The integrator, by design, sees only that card. This provider-versus-integrator information boundary is essential: giving a method the hidden sample identities, generating parameters or white-box graph predictions would change I and invalidate the intended inverse reconstruction comparison.

Note that I1 discloses provider-level regions A_i, not the graph-level SLA query region A_G for the composed service. For each experimental comparison, A_G is fixed and shared by the methods being evaluated.

3.2 Enriched I2 interfaces

The post-result I1/I2 experiments were developed in the repository's `first_science/phase6/` folder (historical Phase-6) and belong to global Phase III in this report. They retain the I1 provider regions and horizon grid and reuse the corresponding provider-local simulation evidence. What changes is the information disclosed to the integrator about how often, and how consistently, each provider satisfies its advertised regions.

I2a: distribution of compliance across simulation runs. Fix one advertised provider region and one evaluation horizon. In each independent simulation run, the provider handles many requests. We calculate the percentage of requests decided by that horizon that satisfy the region's latency, cost and quality limits. This gives **one compliance percentage per run**. To illustrate, suppose four runs produce 98%, 96%, 94% and 70% compliance. If the SLA requires at least 95%, two runs satisfy it and two do not. For this threshold, I1 would report only the pass rate, 2 out of 4 runs (50%). I2a would instead disclose the four compliance percentages themselves, sorted as 70%, 94%, 96% and 98%. The integrator can therefore see not only how often the SLA is met, but also how much compliance varies and how severely it can fall below the requirement. In the actual experiment, **100 independent runs**, rather than four, provide 100 compliance percentages for each advertised region and horizon. These percentages are sorted separately for every horizon, so I2a does not reveal which values at different horizons came from the same run.

I2b: paired compliance at two horizons. I2b includes the I2a distributions and adds pairs of compliance fractions measured in the *same* trajectory at two selected horizons. A pair such as (0.98, 0.91) means that cumulative compliance in one run declined from 98% to 91% between those horizons; the fractions still refer to all requests decided since the start, not just requests processed within the interval. For each selected interval, the published representation contains 100 such pairs. The frozen design samples intervals starting at 5 seconds and then every 10 seconds, with separations of 20, 30, 40, 50, 75, 100, 150 or 200 seconds where both endpoints fit within the 240-second horizon. The pairs reveal whether good or poor compliance tends to persist or change over time. They contain no trajectory identifiers, and the published records cannot be linked across different intervals by a shared identifier; they are not complete request histories or labelled trajectories.

Additional disclosure in I2b-v4. The later I2b-v4 extension publishes one more value for each selected interval: the average, across the 100 runs, of the difference between the highest and lowest cumulative compliance fractions recorded during that interval on the 5-second evaluation grid. This describes variation *inside* an interval that its two endpoint values alone cannot reveal. Individual runs' ranges and complete histories remain private. Section 4.3 explains how M2 uses I1, I2a, I2b and this additional summary to reconstruct provider models. The results appear in Section 9.3.

4. Integration algorithms and historical method versions

4.1 M0: analytical public-boundary predictor

M0 asks a worst-case question: **If every provider meets its advertised limits for a request, what limits follow for the complete service?** The advertised region $A_i(\rho_r)$ (Eq. (5)) specifies maximum latency ℓ_i, maximum cost c_i and minimum quality q_i. In a sequence, latencies add. In an all-required parallel join whose branches start together, latency is determined by the slowest branch. Cost adds because every required service incurs cost. The model defines composed quality as the minimum across required services. Before including processing and communication overheads, the limits are

$$\begin{aligned} \text{Seq}: &\quad \ell = \sum_i \ell_i, \quad c = \sum_i c_i, \quad q = \min_i q_i, \\ \text{ParAll}: &\quad \ell = \max_i \ell_i, \quad c = \sum_i c_i, \quad q = \min_i q_i. \end{aligned} \tag{7}$$

These limits apply to a request only if all required providers respect their advertised limits. M0 applies the same rules recursively to graphs built from sequence and all-required parallel operations. For example, in $G = A \to (B \parallel C)$, the provider-only latency limit is $\ell_A + \max(\ell_B, \ell_C)$, cost is $c_A + c_B + c_C$, and quality is $\min(q_A, q_B, q_C)$. Fixed processing, network and synchronization costs or delays are included at their actual positions in the graph.

For the frozen Phase-I pilot, Source $\to$ Fpre $\to (A \parallel B \parallel C) \to$ Fpost, three network-data hops contribute 0.001001 s each, a completion-control message contributes 0.001 s, and Fpre and Fpost contribute 0.005 s each. Thus the additional latency is $3(0.001001) + 0.001 + 2(0.005) = 0.014003$ s. The two fixed services add $0.015 + 0.015 = 0.03$ cost units, while network costs are zero. The resulting limits are $\ell_G^{M0} = 0.014003 + \max_i \ell_i$, $c_G^{M0} = 0.03 + \sum_i c_i$ and $q_G^{M0} = \min_i q_i$. These constants come from the simulator configuration and must be recalculated for other graph configurations.

Applying these rules to the advertised I1 regions gives the M0-induced region A_G^{I1+M0}, which need not equal the requested graph-level SLA region A_G. For example, M0 may calculate an 80 ms latency limit while the requested SLA allows 100 ms. M0 makes a prediction only when $A_G^{I1+M0} \subseteq A_G$: its calculated maximum latency and cost do not exceed the requested maxima, and its minimum quality meets the requested minimum. Otherwise it reports **not applicable**, which does not mean the service will fail. Provided the required I1 values are available, the prediction is

$$\widehat{\sigma}_G^{M0}(A_G, H; \rho) = \prod_{i \in \{A,B,C\}} \widehat{\sigma}_i^{I1}(A_i(\rho), H; \rho). \tag{8}$$

M0 estimates the SLA probability from the providers' I1 cards; the white-box reference evaluates the requested graph-level region A_G directly. With fixed cards, H and ρ, M0 gives the same probability for every requested region containing A_G^{I1+M0}, even though the white-box probability can change with A_G. The estimate can also be wrong when the regions coincide: providers can fail on different requests, and the product in Eq. (8) does not account for those different failure sets or for execution coupling.

4.2 M1: one reconstructed provider process per provider

M1–M3 reconstruct stochastic provider-process candidates from the disclosed evidence. The assumed process family for each provider is parameterized by

$$\theta_i = (\mu_i, \kappa_i, \mathrm{CV}_i), \tag{9}$$

where μ_i is mean service time, κ_i is cost rate and CV_i is the coefficient of variation of service time. The actual generating parameters are hidden from the integrator; fitted candidates need not recover the physical truth.

M1 fits one stochastic surrogate to each provider's public evidence, combines the three selected provider models and simulates the native graph. The original Phase-3/4 method used a larger single TPE search with its own confirm/replay process and selected one provider hypothesis each. The executed Phase-5 V3b version instead started from **seven independent 25-trial TPE searches per provider**, with separate local search seeds. It rescored the seven run winners on a fresh common $N = 100$ local evidence bank and selected the minimum I1 RMSE candidate. A second independent local bank was diagnostic and not used to choose the model.

A reconstructed parameter vector is a **candidate fitted to I1**, not a recovered physical truth. Later privileged P2 tests found lower I1 errors at the actual generating parameters than at the selected candidates.

4.3 M2: equal-weight finite model propagation

Given three candidate provider models for each provider, M2 propagates the $3^3 = 27$ Cartesian joint models. Its point estimate is

$$\widehat{\sigma}_G^{M2} = \frac{1}{27} \sum_{j=1}^{27} \widehat{\sigma}_G(A, H; \rho \mid \vartheta_j). \tag{10}$$

The historical **Phase-4 M2** selected deliberately diverse I1-compatible representatives and **excluded the M1 anchor** from its portfolio by contract. The executed **Phase-5 V3b M2** chose ranks 1–3 by common-bank I1 RMSE among seven candidates per provider, so the M1 rank-1 provider model **is included** in each V3b M2 local portfolio. There was no hard I1-compatible threshold in the V3b ranking. M2's reported 27 models are thus not a fixed scientific intervention across phases.

The main Phase-5 graph budget for M2 was 100 trajectories per joint model, **2,700 trajectories per physical cell**. Shared seed indices were used across M2 member graph simulations, so a paired seed-index bootstrap is the appropriate numerical dependence-aware design.

Phase-6 M2 reconstruction variants. The six evaluated variants retain the 27 equally weighted joint models of Eq. (10), the same provider-model family and parameter domains, and matched reconstruction and graph-simulation budgets. Each provider search evaluates 256 candidate parameter settings (64 initial space-filling proposals, then 192 adaptive proposals). Each candidate is initially scored using 25 provider simulation runs; the best 24 are rescored with a separate common bank of 100 runs, and the best three per provider are retained. Each of the resulting 27 joint models is simulated 100 times in the graph, for 2,700 graph runs per variant. Search settings and random-number banks are matched, although adaptive proposals can differ when the fitting scores differ. The variants differ in **which disclosed evidence they use and how they select candidate provider models**:

M2 variant	Public evidence and model-fitting criterion
I1STRONG_M2	Uses the I1 probability surfaces. It selects candidate models that best reproduce the published probabilities of meeting each specified compliance threshold, across regions and horizons.

M2 variant	Public evidence and model-fitting criterion
I2AFS_M2	Uses the I2a distributions of per-run compliance. It selects candidates by comparing their predicted and disclosed distributions across all possible compliance thresholds, rather than only I1's five thresholds.
I2B_M2	Uses the I2b pairs of compliance fractions at two horizons in the same run. It scores how closely the simulated *distribution of pairs* matches the published one, giving equal overall weight to each of the eight time separations and equal weight to windows within each separation.
I2BVAR_M2	Uses the same I2b pairs and pair-matching score, but weights each window by the average squared change in compliance between its endpoints. The weights are normalized over all windows, so regions and time separations can receive different overall importance.
I2BHIER_M2	Uses the same I2b pairs and pair-matching score. It preserves equal total weight for each of the five regions and eight time separations, while giving more weight to starting horizons with larger endpoint changes within each region and separation.
I2BV4_M2	Uses the I2b endpoint pairs plus the new public mean within-interval range from Section 3.2. The range determines the weights of endpoint-pair matching within each region and separation; candidate models are not directly scored against the additional range statistic.

Thus **I2a and I2b specify what providers disclose**, while these six M2 variants specify how a method fits provider models using that information. I2BVAR_M2 and I2BHIER_M2 change only the fitting weights applied to the same I2b pair data. I2BV4_M2 also uses an additional disclosed statistic. Comparisons across interfaces therefore test implemented disclosure-and-fitting combinations, not the isolated benefit of more information. All these Phase-6 tests are post-result development evidence; Section 9.3 reports their outcomes.

4.4 M3: public-evidence-weighted finite reconstruction

M3 forms a provider-model candidate bank and scores its members only against public information. In the evaluated family, finite provider weights take the form

$$w_{i,k} = \frac{\exp[-\lambda E_{i,k}]}{\sum_{k'} \exp[-\lambda E_{i,k'}]}, \tag{11}$$

where $E_{i,k}$ is a public-I1 loss (for the executed version, a Jeffreys-smoothed Bernoulli KL discrepancy) and the temperature parameter λ was selected from public-evidence development procedures. A joint model $\vartheta_j = (\theta_{A,a}, \theta_{B,b}, \theta_{C,c})$ receives product weight

$$W_{a,b,c} = w_{A,a} w_{B,b} w_{C,c}. \tag{12}$$

This is a weight over *hypotheses about providers*. The product does not remove the actual queueing/dependency interactions once a joint triple is propagated through the native simulator; those interactions remain inside each stochastic graph run. These Gibbs weights are **not calibrated posterior probabilities** of the hidden generating model.

For a selected set S, normalized aggregate weights $\alpha_j = W_j / \sum_{k \in S} W_k$ yield

$$\widehat{\sigma}_G^{M3} = \sum_{j \in S} \alpha_j \widehat{\sigma}_G(A, H; \rho \mid \vartheta_j). \tag{13}$$

The **historical pilot M3-v4** began with **48 candidates per provider**, used public-I1 Gibbs weights with $\lambda = 30$, and propagated the **Top14** joint hypotheses under total graph budgets $B = 1400$ or 2000. Within this *finite* inferred distribution, those 14 members carried approximately 99.917% of the joint weight.

The executed **Phase-5 V3b M3 FULL343** instead used **seven independently obtained candidates per provider** and propagated **all** $7^3 = 343$ **joint hypotheses** with total budget $B = 1400$. This replaced the registered Top14 execution after a **pre-graph-white-box** mass diagnostic found Top14 retained mass only 0.552 in P1, 0.962 in P3, and 0.867 in P4. It preserved the candidate bank and frozen weights, assigned at least one graph trajectory per

joint member and greedily allocated the remaining budget using the reduction in the upper variance objective. The change was documented before the end-to-end graph reference was read, but it was still a **method revision**, not a confirmatory unchanged test of originally registered Top14.

For weights W_j and independent Bernoulli member trajectories,

$$\mathrm{Var}_{MC}(\hat{\sigma}_G^{M3}) \le \frac{1}{4}\sum_j \frac{W_j^2}{N_j}, \qquad \sum_j N_j = B. \tag{14}$$

The continuous relaxation gives $N_j \propto W_j$; the finite algorithm initializes one trajectory/member and iteratively chooses the largest $\Delta_j = W_j^2/[N_j(N_j+1)]$, with a frozen rank tie-break rule. Archive verification reconstructed the P1/P3/P4 allocations **exactly**.

Crucially, **M3 Top1 is not definitionally M1**. M1 selects per-provider candidates by minimum common-bank I1 RMSE. M3 Top1 takes the maximum *product* of provider Gibbs weights based on KL energies; the two may select the same parameter tuple in some worlds, but identity is not guaranteed.

4.5 Within-ensemble support bounds

For any finite normalized Gibbs support, let S retain weight mass $m_S \in (0,1]$, and let p_{full} and p_S be respectively the expected SLA probability under the full finite distribution and the normalized retained submixture. Since each member probability is in $[0,1]$,

$$\boxed{|p_{\text{full}} - p_S| \le 1 - m_S.} \tag{15}$$

The proof writes $p_{\text{full}} = m_S p_S + (1-m_S)p_{\text{omitted}}$ and uses both probabilities in $[0,1]$. This is an **exact within-bank truncation bound**. For the pilot Top14, $1-m_S$ was approximately 0.000826; for Phase-5 FULL343 it is zero **relative to the chosen 343-member bank**. Neither statement bounds error with respect to hidden graph truth.

Method-version matrix

Historical Phase-4, Phase-5 and Phase-6 names identify repository experiment stages, **not** further overarching research phases. They belong respectively to global Phases I, II and III.

Period	M1 selection	M2 membership	M3 support	Main M3 graph budget	Evidence role
Phase-4 pilot	Historical TPE single fit/provider	3 diverse/provider; M1 anchor excluded	48/provider → weighted Top14	1,400 or 2,000	Same-world frozen pilot
Phase-5 registered V2	Frozen TPE/domain policies	Separate diversity/compatibility policy	Originally Top14	Planned 1,400	Protocol, not final executed version
Phase-5 executed V3b	Best common-$N100$ RMSE of seven	RMSE ranks 1–3; includes M1 anchor	All 7/provider → FULL343	1,400	12-cell final-WB evaluation
October retrospective M3	V3b remains comparator	Not re-estimated	48/provider Top14 and Top1	1,400 each	Post-result same-WB comparison
Phase-6 matched	Strong I1 or I2-specific loss and 256-trial search	Top3/provider, 27 equal weights	**Not evaluated**	Not applicable	Post-Phase5 development

The matrix deliberately does **not** imply identical candidate search, local costs, physical-cell populations or empirical credibility across the rows.

5. Native simulator, graph structures, and experiment protocol

5.1 Modified YAFS/AICon composition engine

The discrete-event environment uses YAFS as extended under AICon [9,11]. Modified YAFS application/core code registers compositions and per-branch dependencies, carries a per-request **composition path**, activates a dependent branch when its required sibling completions for that invocation have been received, and emits a single all-required composition output. Standalone code probes cover elementary joins, scoped dependency isolation and nested compositions.

The Phase-5 scientific test does **not** exercise arbitrary nested service DAGs. It compiles a small family of exactly three-provider sequence/parallel ASTs into dependencies in **one** native composition controller. Source/Fpre/provider/Fpost execution uses fixed physical node embedding and placement across all four logical graphs. Periodic arrivals are sent to stochastic provider services, with native queueing, service cost and network delays.

The four frozen dependency maps are:

Logical graph	A predecessors	B predecessors	C predecessors	Reading
G_PAR	none	none	none	$A \parallel B \parallel C$
G_SEQ	none	A	B	$A \to B \to C$
G_SEQPAR	none	A	A	$A \to (B \parallel C)$
G_PARSEQ	none	none	B	$A \parallel (B \to C)$

These are four logical composition structures, not four separately changed physical infrastructure topologies. The companion source-level audit documents the Graph AST compiler, validation of unique leaves, deterministic replay probes and controller source locations.

The inspected runtime constructs an empty acting management-agent network. It therefore evaluates *offline graph performance estimation*, not online active-inference orchestration, decentralized planning or adaptive placement. WATT is zero in these test topologies and messages use a neutral QoS propagation specification; although (L, C, Q) is the formal admissibility schema, strong empirical energy/quality variation was not established by the core battery. These are limits on external validity, not omissions to conceal.

5.2 White-box worlds, public evidence, and test set design

The research stages intentionally distinguish **disclosure production**, **query design**, **reconstruction**, **blind prediction**, **final white-box reference** and **evaluation**. The ability to inspect generating provider parameters belongs to privileged experiment construction and later retrospective diagnostics, not to the permitted reconstruction method.

The Phase-5 registered experiment included four provider worlds P1–P4, the four graph forms above and a five-ρ by three-regime query battery in each planned world-by-graph cell. Step-0 used dedicated calibration and confirmation white-box evidence to freeze nondegenerate regional difficulty and determine eligibility. This evidence was allowed for **experiment design only**; it was not the final embargoed graph reference. Four P2 cells failed the registered Step-0 gate. They were recorded as missing/failed design conditions and **not repaired, silently replaced or included** in the 12-cell evaluation.

The main evaluation used horizons $H = 60, 65, \ldots, 240$ seconds, **37 per query**, 15 query conditions per eligible physical cell, or **555 related SLA measurements per cell**. With 12 eligible cells, this yields 6,660 points per fully applicable M1/M2/M3 method. Each cell's separate final graph white-box reference used 1,000 trajectories, held back until global prediction freezing.

5.3 Execution chronology and integrity controls

The registered research pipeline was intended to flow from engineering smoke through I1 publication, Step-0 eligibility, reconstruction, blind graph prediction, global freeze, final WB and deterministic evaluation. The Phase-5 V3/V3b and FULL343 revisions occurred after examining specific *permitted* local/mass diagnostics but before the final graph WB was opened. This keeps an important end-to-end reference separation, **without** pretending the entire original V2 method was held constant.

The user-executed local B1 provenance audit found **94 passed checks, no failures or warnings, five informational statuses**. In particular, all 12 blind-prediction cell hashes matched the global freeze manifest, and all 12 final-WB cell manifests linked to the same freeze. The global prediction freeze was recorded on **7 October 2026 at 19:26:06 UTC**, before the earliest final white-box cell start at **19:51:50 UTC**, a gap of **25 minutes 43 seconds**. These are independently checked archived-manifest links and timestamps, not a formal information-flow proof.

The required engineering-smoke checks were ultimately completed, with four frozen PASS manifests marked **non-scientific**. Nevertheless, some mandatory supplemental checks were performed **after** early I1/Step0/M1/M2 scientific activity. This is a documented **sequencing deviation**, which later smoke completion does not retroactively erase. The repository separately records source commits, SHA-256 manifests and seed ranges. The external AICon/YAFS runtime currently resolves to the public September 2 commit, but a contemporaneous pristine imported YAFS build for every October execution has not been attested. Exact bitwise re-execution is therefore not claimed.

6. Research sequence and comparison validity

The three overarching phases below are separate from the original experiment-stage labels. Historical labels are preserved to identify frozen methods and archived evidence; exact paths and freeze IDs appear in Appendix B.

Overarching phase	Historical stage	Evidence role
I: Controlled pilot	Historical Phases 1–3 and Phase-4 G1/G2	Simulator, I1 disclosure, M0/M1/M2 development
I: Controlled pilot	September M3 Top14 pilot, N200	Frozen same-condition reference evaluation
I: Controlled pilot	September Top1/Top3 and hidden-parameter audits	Post-hoc pilot diagnostics, no new holdout
II: Expanded battery	Historical Phase-5 V2	Registered 4-world/4-graph design
II: Expanded battery	Historical Phase-5 V3/V3b and FULL343	Revised methodology before final WB
II: Expanded battery	Historical Phase-5 final	12 eligible cases, N1000 reference
III: Retrospective divergence tracing	October original-style M3 Top14/Top1	Post-result comparison on Phase II reference
III: Retrospective divergence tracing	D0–D9 pilot–battery controls	Forensic parity and protocol audit
III: Retrospective divergence tracing	Historical Phase-6 I1/I2 and October P2	Related post-result development, separate tests

The global phases classify research purpose and evidence status, not three unchanged algorithms. Phase II's documented method revisions occurred before its final white-box reference but differed from the initially registered method. Phase III results are retrospective or exploratory. Cross-phase comparisons require comparable populations, methods and budgets.

7. Phase I: controlled pilot

7.1 Query population and reference

The original pilot held one hidden three-provider configuration, one fixed parallel-all graph, a workload period of 0.2 seconds and the providers' published I1 cards constant. Query difficulty was varied by scaling the latency and cost limits of a graph region induced from the providers' advertised regions:

$$A_G(\gamma, \rho) = \{L_G \le \gamma \ell_G^{\text{base}}(\rho),\ C_G \le \gamma c_G^{\text{base}}(\rho),\ Q_G \ge q_G^{\text{base}}(\rho)\}. \tag{16}$$

Here $A_G^{\text{base}}(\rho)$ is the region A_G^{I1+M0} calculated by M0 from the advertised provider limits. The scaling factor γ changes the requested region scored by the white-box reference, but does not change the I1 provider probabilities used in M0's product. In particular, if $\gamma > 1$, the requested latency and cost limits are looser than M0's calculated limits. Thus the white-box reference is evaluated at the larger requested region, whereas M0 still uses the same provider-card product. If $\gamma < 1$, M0's calculated limits do not fit the tighter query and M0 is not applicable.

Five thresholds times three difficulty regimes generated 15 queries. Evaluation used 37 horizons from 60 to 240 seconds. The pilot's fresh reference bank contained $N_{\text{WB}} = 200$ trajectories.

7.2 Prediction errors and historical cutoff diagnostic

The original pilot production results were re-scored from saved pointwise tables and manifest-linked hashes:

Method, pilot	Graph trajectories	Applicable points	MAE	RMSE	Signed bias
M0, applicable subset	0	333	0.57204	not included in principal comparison	-0.57204
M1	100	555	0.18517	0.23074	+0.18517
M2, 27 equal members	2,700	555	0.06815	0.09414	+0.04737
M3 pilot Top14, $B = 1400$	1,400	555	**0.03037**	0.03578	+0.02824
M3 pilot Top14, $B = 2000$	2,000	555	**0.02755**	0.03306	+0.02447

MAE, RMSE and bias are computed against the finite WB reference. Positive bias means optimistic prediction. M0 has a different applicable query set (9/15 queries, 333 horizon points), so its row must not be ranked directly against all 555-point methods. The main M3 graph-propagation budget B=1400 is 48.15% smaller than M2's 2700 trajectories, not proof of smaller end-to-end computational cost including provider fitting. The M0 RMSE value is deliberately not invented where the archived review does not report it as a reconciled principal comparison.

As a **secondary historical diagnostic**, the pilot applied an arbitrary cutoff $\beta = 0.90$ to predicted and white-box-referenced SLA-satisfaction probabilities and compared the resulting yes/no labels. This cutoff was **not a contractual SLA requirement**, and the diagnostic was not part of the main probability-estimation objective. The audit used reference-WB pointwise Wilson intervals to identify **481 reference-certain** query/horizon points among 555. Only **274** were M0-applicable and reference-certain. The historically observed agreements were **33.94%** (M0 on 274), **37.63%** (M1), **86.07%** (M2), **96.47%** (M3 B1400), and **97.09%** (M3 B2000), with the latter four computed on 481 points. These percentages are **descriptive conditional agreements against a finite WB reference**, not calibrated real-world admission success rates or independent-scenario accuracy estimates.

The pilot's M3 production errors were substantially smaller than those of M1/M2 in this exact world and graph. That is a genuine, archived numerical observation. It does not demonstrate the same numerical advantage in worlds with altered provider demand or different logical dependencies.

7.3 Retrospective support and weighting

A later frozen pilot Stage-B audit compared several estimators at matched total $B = 1400$ in the same world:

Retrospective pilot estimator	MAE	RMSE	Bias
Top1 B1400	**0.02878**	0.03406	+0.02498
Uniform14 equal-allocation B1400	0.05490	0.06570	+0.05362
Weighted14 equal-allocation B1400	0.04222	0.05016	+0.04066
Weighted14 production-allocation B1400	0.03037	0.03578	+0.02824

Using the **same 14 model members and the same equal-allocation ledgers**, weighted versus uniform averaging is the cleanest isolated *weighting* contrast. However, the production versus equal-allocation comparison also changes seed/trajectory allocation; its difference is descriptive rather than a clean causal isolate. Top1 does slightly better than production weighted14 on this same retrospective fixed-world matched-budget test, which defeats any claim that *retaining more models is intrinsically necessary for point accuracy*.

The historical operational readouts also reported Top1 B100 and Top3 B300 MAEs near 0.0399 and 0.0401, respectively, but the direct underlying first-100 source arrays have not yet been independently reconciled in the B1 checks. They remain **reported historical numbers, not upgraded to primary independently checked evidence**.

7.4 Hidden-parameter audit and restricted cost frontier

The retrospective pilot Table 4 compared hidden generating parameters with M1 and M3-Top1 surrogate parameters and public-I1 reconstruction losses. The archived values were independently reconciled. Their discrepancies, including notably different Provider C reconstructions, show that **good graph prediction can co-occur with very poor recovery of hidden parameters in this one pilot**. They do not prove the public interface cannot identify those parameters in principle.

The historical Figure 2 labeled a curve the "nondominated" recurring-cost/MAE frontier while plotting only chosen primary prospective implementations. In the same reported two-coordinate cost plane, the retrospective operational Top1 at B100/MAE 0.0399 dominates M1 at B100/MAE 0.1852. Thus the original line is **not** the nondominated frontier over all reported operating points; it can only be described as the frontier of the *selected prospective primary set*, or it must be redrawn and annotated with retrospective status and additional provider-construction cost. This first B1 draft does not reuse the misleading frontier graphic.

8. Phase II: expanded battery

8.1 Eligibility and exact prediction population

Phase-5 planned 16 physical world-by-graph cells. Under the Step-0 eligibility/confirmation protocol, the four cells in world P2 failed. The remaining **12** cells cover worlds P1, P3, P4

across G_PAR, G_SEQ, G_SEQPAR and G_PARSEQ. Each has 15 frozen SLA queries, evaluated at 37 horizons; each has a distinct fresh $N_{\mathrm{WB}} = 1000$ graph reference.

There is **no estimate for the preregistered P2 component of the Phase-5 directional heterogeneity contrast**. The later induced-public-region P2 study used a different query design and cannot be substituted into this missing prospective cohort.

8.2 Executed V3b method comparison

The Phase-5 evaluation used V3b-selected candidate banks: **M1 = one joint model, M2 = 27 equal-weight joint members, M3 = FULL343 weighted joint members**. The independently checked saved pointwise metrics are:

Phase-5 executed V3b	Evaluated points	Graph trajectories/cell	MAE	RMSE	Signed bias
M0 on applicable subset	1,776	0	0.41874	0.45392	-0.41808
M1	6,660	100	0.15702	0.22588	+0.14968
M2	6,660	2,700	0.13395	0.19045	+0.10094
M3 FULL343	**6,660**	**1,400**	**0.12189**	**0.18081**	**+0.06511**

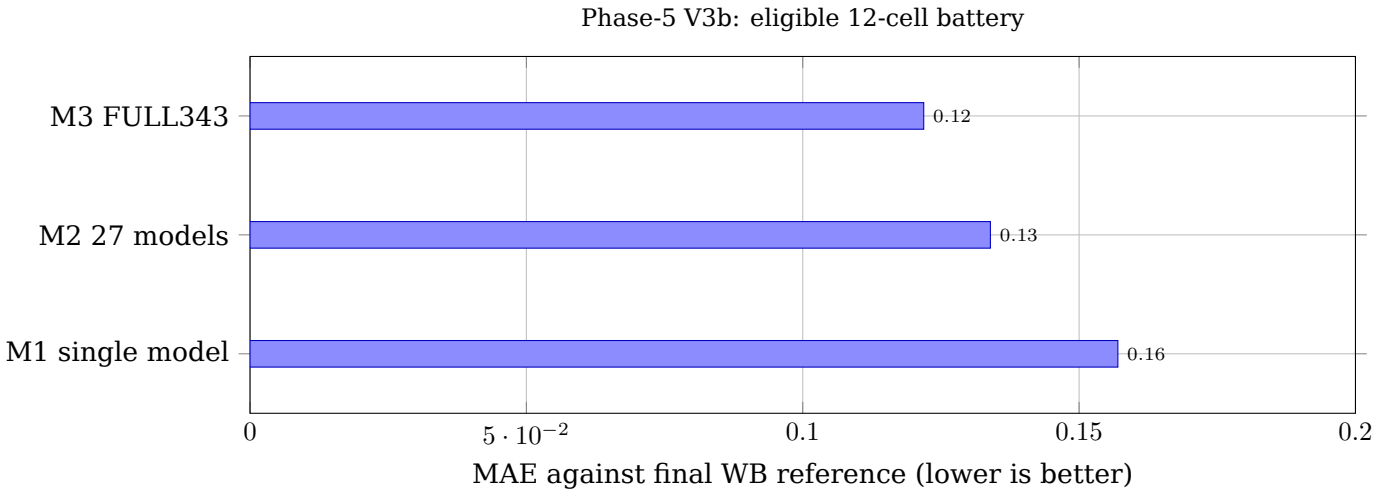


Figure 1: Phase-5 aggregate MAE from verified frozen V3b pointwise results. The full 12-cell eligible battery contains 6,660 related SLA measurements (across conditions and time horizons) per model. Different graph budgets apply: M1 100, M2 2,700 and M3 1,400 trajectories per cell. M0 is omitted because its applicable subset is different.

These are aggregate scores over the **12 eligible physical cells only**, not a population-generalization estimate. No method dominates all conditions. The remaining MAEs, even for M3, are substantial relative to probabilities bounded by one. The positive aggregate biases for reconstructed methods indicate a systematic tendency towards overprediction within this eligible design, though their magnitude varies with the condition.

Because M0 is defined on a smaller and selected applicable set, its aggregate MAE cannot be directly compared to the all-query rows. Moreover M1/M2/M3 have sharply different graph simulation budgets, so the aggregate ranking does not isolate simulation efficiency at equal cost, nor does the pre-WB change from Top14 to FULL343 permit a direct unchanged-algorithm replication of the September pilot.

8.3 Topology and provider-world effects

The prospectively specified topology statistic for each method and provider world was

$$\Delta_P^M = \mathrm{MAE}_M(P, G_{\mathrm{SEQ}}) - \mathrm{MAE}_M(P, G_{\mathrm{PAR}}). \tag{17}$$

The original directional masking hypothesis anticipated $\Delta_P^M > 0$. The completed analysis did **not** yield that sign consistently across the eligible worlds: P1 differed in sign from P3/P4 for the examined contrast. It is therefore not supported as a universal or robust rule by this evaluation. A second registered heterogeneous-minus-symmetric contrast required P2 and was **not estimable** after Step-0 exclusion. The two other graph ASTs offer mechanistic triangulation but do not constitute a clean factorial manipulation separating workload shift from aggregation structure.

The 6,660 measurements reuse simulation trajectories across SLA conditions and time horizons, so they are not independent experiments. They cannot be treated as 6,660 independent physical deployments when assessing variation across provider configurations. A WB-reference bootstrap resamples the **complete trajectory/seed index** jointly across horizons, queries and shared-seed cells, keeping method predictions fixed. It addresses finite-N_{WB} reference fluctuation, **not model-selection uncertainty or future-world robustness**.

The scientifically material result is twofold: a weighted finite ensemble maintained a favorable aggregate *ranking* among the executed methods, and its **absolute prediction remained too inaccurate in several difficult regions** to claim dependable generic SLA estimation.

9. Phase III: retrospective divergence tracing

Phase III brings together original-method comparisons on the previously inspected Phase II reference, D0–D9 pilot–battery forensic checks, and related post-result exploratory investigations. The latter retain their historical experiment labels and do not provide a new independent holdout.

9.1 Original-style M3 transfer to the Phase II battery

To address the missing comparison with original M3, a **retrospective** experiment reconstructed the registered Phase-5 V2-style M3 method across all **12 Step-0-eligible physical cases** (P1, P3 and P4, each on four graph structures). It was designed **after** earlier Phase-5 V3b reference outcomes had been inspected. It reused the frozen 15 SLA queries per cell, 37 scored horizons (60 to 240 seconds), and N=1000 final white-box reference per cell. P2 was neither repaired nor reintroduced.

For each provider in each eligible world, the registered V2-style **48-point nonadaptive LHS** was sampled within the frozen search domain and scored using **public I1** and the frozen N=100 local-rescore bank. Jeffreys-smoothed Bernoulli-KL with $\lambda = 30$ gave normalized provider Gibbs weights whose products defined $48^3 = 110{,}592$ joint models. The 14 highest-weight combinations were propagated with **B=1400 graph trajectories per case**, allocated by the historical integer minimax variance-bound rule. The highest-weight **Top1 was separately simulated at B=1400**. All 12 retrospective predictions were frozen before this retrospective code read the existing white-box result. The seven-fit **V3b FULL343** remains an unchanged comparator. This reproduces key registered V2 M3-style rules, but is **not** a bit-for-bit reproduction of the September Phase-4 pilot.

Method on 12 eligible cells	Points	MAE	RMSE	Bias
Executed V3b FULL343	6,660	**0.121890**	**0.180806**	+0.065105
Retrospective original Top14	6,660	0.154155	0.212843	+0.044867
Retrospective original Top1	6,660	0.162280	0.225373	+0.054562

Top14 retained mass within the reconstructed 48-per-provider bank was **0.996404 (P1)**, **0.996771 (P3)** and **0.998925 (P4)**. Section 4.5 therefore bounds the *full-bank versus*

truncated-bank expectation discrepancy by 0.003596, 0.003229 and 0.001075, respectively. This bound does **not** cover graph Monte Carlo noise or guarantee accuracy against the true composition. High internal support coverage coexists with material external prediction error.

Provider world	FULL343 MAE	Top14 MAE	Top1 MAE
P1	0.17326	0.21172	0.21149
P3	0.07169	0.12659	0.14299
P4	0.12073	0.12415	0.13237

Graph	FULL343 MAE	Top14 MAE	Top1 MAE
G_PAR	0.13586	0.20183	0.21773
G_PARSEQ	0.11016	0.14450	0.14318
G_SEQ	0.14624	0.14642	0.16105
G_SEQPAR	0.09530	0.12387	0.12716

FULL343 achieves lower MAE than original-style Top14 in **11 of the 12 cells**; Top14 beats its separately simulated Top1 in **8 of 12**. The advantage is heterogeneous. **P3/G_PAR contributes 49.3%** of the aggregate Top14-minus-FULL343 MAE gap (0.25108 versus 0.06001), while **P3/G_SEQ reverses the ranking** (0.08302 for Top14 versus 0.11633 for FULL343). Removing P3/G_PAR leaves a mean FULL343 advantage of 0.01783 across 11 cells. P4 is almost tied.

These findings **do not establish that larger ensembles cause better predictions**. LHS versus seven-fit V3b changes proposal/fit procedure, selection and weights; **equal graph budgets do not imply equal total reconstruction cost**. The 6,660 query-horizon points are related observations, not independent experiments. This is a post-result comparison, not a fresh holdout.

The uploaded archive passed **316 independently recomputed numeric and SHA-256 checks, zero failures**. The review checked 12/12 frozen prediction-manifest links, all included prediction hashes, provider-weight normalization, Top14 members from all 48^3 products, and pointwise/error values. FULL343 reproduces its earlier MAE 0.121890. Missing member-curve files and raw native graph ledgers prevent an independent bitwise simulation replay.

9.2 Pilot–battery divergence tracing

This Phase III forensic control tests what was preserved when progressing from the Phase I controlled pilot to the Phase II expanded battery. It uses archived records and sampled replay trajectories, not a new prospective accuracy test.

Estimator and simulator equivalence. The pilot's provider Gibbs weight vectors were reproduced by the newer helper to within 2.78×10^{-16}, with the same 14 joint members, weights, retained mass (0.999174197017) and B1400 allocations. On archived-seed subsets, G_PAR and historical G0 native request ledgers agree to floating-point precision when both use provider IPT 10^6. Ranks 1–9 used two seeds each (2,398 requests per rank); ranks 10–14 had one available frozen B2000 seed each (1,199 requests per rank). Switching provider IPT to 10^9 introduced small continuous latency/cost differences without changing completion status. Ranks 10–14 also showed **zero changes over 555 original-pilot query–horizon SLA predictions** at either gauge. The complete B1400 experiment and the aggregate 14-rank SLA scoring were not replayed.

Reconstruction-policy deviation. All nine new provider-world combinations reused the original 48 normalized Latin-hypercube coordinates, but different physical parameter-domain maps were used, and every Top1 candidate index changed. The original pilot chose $\lambda = 30$ by five-fold blocked CV on public I1; the broad battery and retrospective instead inherited the fixed value 30. Repeating the same local public-I1-only CV rule retrospectively gives:

World	Best λ	Used λ	Best KL	KL at 30	Excess
Pilot	30	30	0.126042	0.126042	0%
P1	3	30	0.099806	0.124984	+25.23%
P3	5	30	0.082610	0.158658	+92.06%
P4	20	30	0.090682	0.098781	+8.93%

These are **held-out KL scores on the individual providers' advertised probability surfaces, not graph prediction errors**. Different λ would alter propagated model support. A lower CV-KL therefore does *not* demonstrate better global SLA prediction. The pilot's adaptive 99.9%-mass policy yielded K=14; on the new banks it would select K=19 (P1), 22 (P3), or 15 (P4), rather than the fixed K=14.

Provider-level versus composed-graph fit. Mean provider-level Full48 public-I1 MAE was 0.1277 (pilot), 0.1314 (P1), 0.1556 (P3) and 0.1101 (P4). Corresponding original-style Top14 G_PAR graph MAEs were 0.0304, 0.2447, 0.2511 and 0.1098. The two error measures have different targets and cannot be equated, but the pattern rejects a simple account of uniformly worse provider-level fit. Sensitivity to particular SLA queries, approximation errors and error cancellation remain possible explanations; inherent insufficiency of I1 is *not* proved.

Two interacting methodological challenges emerge: **predictive sufficiency** of provider statistics reconstructed from individual disclosures for composed SLA events, and **experimental-protocol fidelity** as parameter domains, hyperparameter selection, support rules and query regions change. D0–D9 close a bounded forensic audit; they do not uniquely attribute the pilot–battery accuracy gap. A new prospective one-axis-at-a-time experiment would be needed for causal attribution. Full register: `first_science/b1/retrospective_m3/PHASE_III_DIVERGENCE_CONT`

9.3 Post-result I1/I2 pipeline development (historical Phase-6)

After the Phase II expanded battery (historical Phase-5), the six M2 variants defined in Section 4.3 were evaluated on **P4/G_SEQPAR**, using 15 queries × 37 horizons = 555 related SLA measurements per variant. Their reconstruction-search architecture and graph-propagation budgets were matched at **B=2,700 graph runs per variant**. Their disclosed evidence and/or provider-fitting criteria differed, as specified in Section 4.3.

P4/G_SEQPAR M2 variant	MAE	RMSE
I1STRONG_M2	**0.09178**	**0.11390**
I2AFS_M2	0.09475	0.11826
I2B_M2	0.10210	0.13617
I2BHIER_M2	0.09429	0.11695
I2BV4_M2	0.10854	0.15354
I2BVAR_M2	0.15731	0.22086

The stronger I1 control performs best or near-best among these **joint information–loss pipeline** variants in this cell. Richer information is not a monotone improvement *under the tested objectives*. This does not show that I2 information is useless, that an ideal I2 estimator cannot improve on I1, or that the comparisons are independent held-out validation: the P4/G_SEQPAR cell had already informed iterative I2 development.

Privileged known-provider/oracle studies and provider-factorial substitutions were additionally reported, but are not treated as deployable I1-only methods or as equivalently independently audited core results.

9.4 Distinct P2 public-induced-region comparison

The October P2/G_PAR study chose the graph admissibility region deterministically from the **same published I1 rectangles** for all three reconstruction methods. It evaluated five

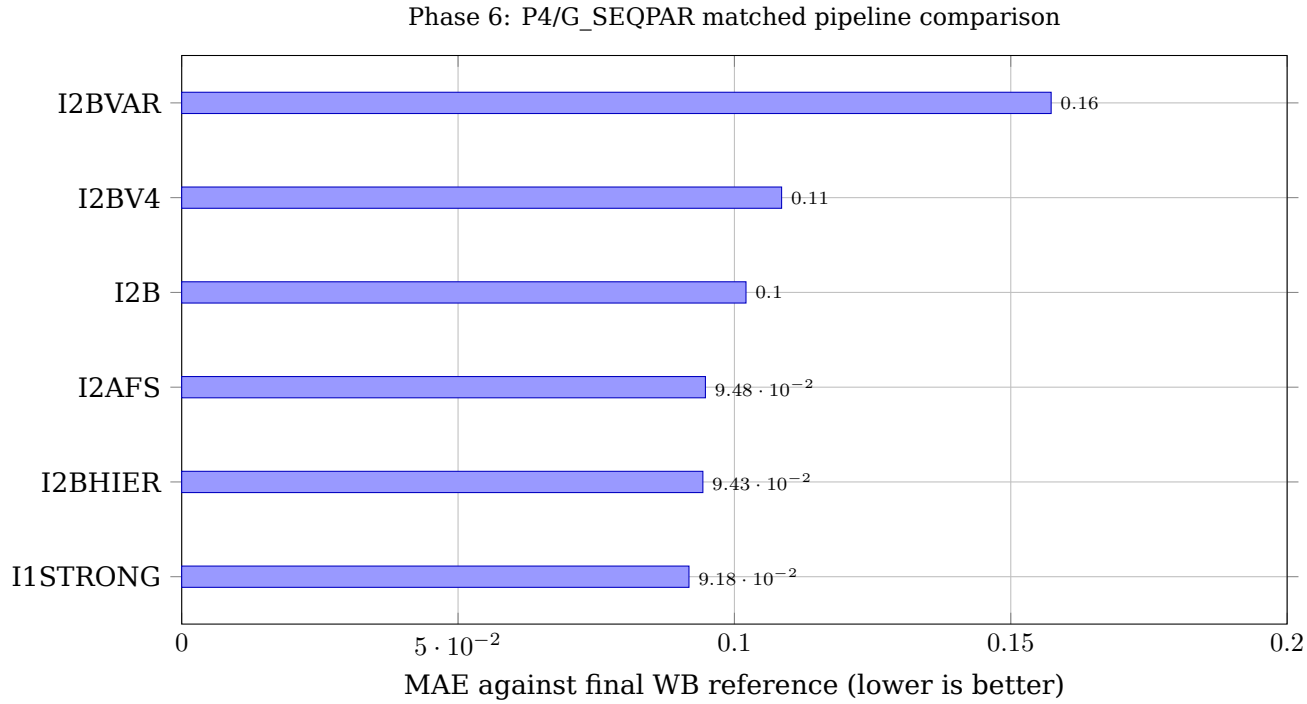


Figure 2: Phase-6 P4/G_SEQPAR matched-budget development comparison. All six rows use 555 related SLA measurements (15 conditions at 37 horizons) and 27-member M2 graph propagation at B=2,700; disclosed information and fitting losses vary together.

same-ρ queries over 37 horizons, or **185 points per method**, with a new reference design distinct from the failed Phase-5 P2 Step-0 conditions.

P2/G_PAR method	Native graph trajectories	MAE	RMSE	Bias
M1 V3b	100	0.13200	0.15332	+0.04149
M2 V3b	2,700	0.14058	0.16371	+0.13791
M3 V3b FULL343	1,400	**0.12787**	**0.14853**	+0.04263

Here all three methods are evaluated against **the same** A_G^{I1+M0}, constructed from the I1 provider limits using M0's boundary rules. They differ in process reconstruction and subsequent graph prediction. Hence no difference above measures graph-region reconstruction quality. The simulation budgets are **unmatched**; M3's small observed MAE advantage does not establish equal-cost superiority. This study cannot be read as a post-hoc repair of four earlier ineligible P2 cells.

9.5 Privileged search-deficit diagnosis

A separate P2 audit evaluated the actual hidden provider parameters against the public I1 target using the **same** $N = 100$ **rescore and independent replay bank** as the finite reconstruction candidates. For each of three providers the true parameter vector was inside the search domain, yet it achieved better I1 RMSE than the selected best available V3b candidates:

P2 provider	True θ rescore RMSE	Selected-best rescore RMSE	True θ replay RMSE	Selected-best replay RMSE
A	**0.03443**	0.10961	**0.03151**	0.12247
B	**0.05337**	0.17780	**0.08178**	0.18344

P2 provider	True θ rescore RMSE	Selected-best rescore RMSE	True θ replay RMSE	Selected-best replay RMSE
C	**0.06110**	0.13099	**0.08437**	0.12106

The registered seven-by-25-trial search generated **525** total provider trials across three providers. If the generating parameters fit the disclosed evidence better than the selected candidates despite being searchable in principle, then the executed **search and ranking procedure failed to find a comparatively favorable fit** under this test. This is a concrete optimization/inference failure mode.

It remains possible for a public interface to be non-identifying in some formal sense; however this particular evidence cannot prove that non-identifiability caused the observed prediction error. A 42-configuration, two-seed-bank $\pm 20\%$ one-at-a-time local perturbation audit additionally measured finite objective sensitivity; it neither identifies the full local Hessian nor proves uniqueness or global optimizer convergence.

10. Uncertainty mathematics and cross-experiment interpretation

10.1 Numerical Monte Carlo precision

For a fixed reconstructed ensemble with normalized weights α_j and estimates $\widehat{p}_j$, let $\widehat{p}_M = \sum_j \alpha_j \widehat{p}_j$. The variance identity is

$$V_{\mathrm{MC}}(\widehat{p}_M) = \sum_j \alpha_j^2 V_j + 2 \sum_{j<k} \alpha_j \alpha_k C_{jk}. \tag{18}$$

The covariance C_{jk} matters for **M2's shared seed indices**. A seed-index paired bootstrap should resample each 27-member seed-vector as one unit rather than resampling individual member trajectories independently. M3's independent member seed streams permit the simpler variance form in Eq. (14), conditional on fixed weights, models and allocation.

The pilot's numerical-width appendix used Wilson intervals for fixed single-model binomial curves, paired percentile resampling for M2 and a normal approximation for M3 using a **worst-case variance**. The variance bound itself is conservative; the resulting **normal-approximation 95% interval is not an exact distribution-free confidence interval**, particularly when the FULL343 allocation leaves some models with only one trajectory. All such intervals are **pointwise** over (A, H, ρ) and conditional on the reconstructed method, not simultaneous across all 555 or 6,660 related SLA measurements.

Separately, the Wilson intervals included in I1 measure sampling uncertainty in the provider-side probability estimates conditional on regions fitted from the independent construction bank. They do not include uncertainty in the fitted regions or provide simultaneous coverage across the I1 probability surface.

A rigorous fixed-sample conditional alternative, if required in later work, can be derived from a weighted Hoeffding inequality for independent bounded member outcomes, but this was **not an evaluated new estimator** in the frozen experiments. Sequentially peeking at ordinary fixed-sample intervals until a threshold is crossed does not preserve nominal coverage without a separately specified anytime-valid procedure.

10.2 Model dispersion is a distinct diagnostic

Let $p_j = E[\widehat{p}_j]$, $p_\alpha = \sum_j \alpha_j p_j$ and $\widehat{p} = \sum_j \alpha_j \widehat{p}_j$. The observed finite-ensemble dispersion is

$$V_{\mathrm{raw}} = \sum_j \alpha_j (\widehat{p}_j - \widehat{p})^2. \tag{19}$$

Under fixed weights and unbiased member estimators, its expectation equals latent finite-model spread plus a contribution from numerical sampling,

$$E[V_{\text{raw}}] = \sum_j \alpha_j (p_j - p_\alpha)^2 + \sum_j \alpha_j \operatorname{Var}(\widehat{p}_j) - \operatorname{Var}(\widehat{p}). \tag{20}$$

The manuscript's nonnegative MC-subtracted plug-in diagnostic estimates only the *spread among the selected models*. The max-with-zero clipping and finite member budgets make it neither an unbiased calibrated risk estimate nor a guaranteed bound on model-to-hidden-world error.

Similarly, if the method predictor and white-box reference are independent and individually unbiased for their conditional targets, then

$$E\big[(\widehat{p}_M - \widehat{p}_{\text{WB}})^2\big] = (p_M - \sigma_G)^2 + V_{\text{MC,M}} + V_{\text{WB}}. \tag{21}$$

Subtracting empirical noise estimates from an observed squared difference and clipping at zero is an **experimental noise-correction diagnostic**. It does not make the unobserved model approximation error known exactly, and its square root adds estimation bias.

The different uncertainties are summarized as follows:

Quantity	Obtained from	What it does *not* certify
MC variance or Wilson/bootstrapped numerical band	Repeated graph simulation of frozen model(s)	Distance to hidden σ_G
Finite-model min/max and weighted spread	Enumerated reconstructed joint hypotheses	Coverage of all physically possible models
Gibbs omitted mass $1 - m_S$	A normalized **chosen finite** Gibbs bank	Truth error outside that bank
Finite-WB reference bootstrap	Reference WB seed/trajectory resampling	Generalization to unseen worlds or new optimization searches
Across-world/graph variation	Distinct designed physical cells	Representative real-world uncertainty from only a few cells
Hidden-truth error and parameter diagnostics	Privileged reference available after freeze	A deployable estimator without privileged evidence

10.3 Cross-phase lessons

These lessons compare global Phase I (controlled pilot), Phase II (expanded battery) and Phase III (retrospective divergence tracing) while preserving their distinct reference populations and evidence status.

Candidate search is a plausible limiting mechanism, demonstrably so in P2. The fact that true parameters better fit I1 than selected candidates makes the inverse search procedure a high-priority diagnosis. It does not eliminate other causes of graph discrepancy, including information compression and model-family restrictions.

Ensembling is neither futile nor proved necessary. The pilot same-support weighting test favored nonuniform public-evidence weights over uniform weights, while pilot Top1 slightly outperformed the larger weighted mixture at a matched budget. In the larger Phase-5 study M3 FULL343 had the best aggregate MAE but still made sizeable errors. The data support context-dependent method performance, not a model-count law.

Internal mixture coverage is not external prediction validity. The retrospective original-style Top14 captured over 99.6% of the inferred joint-model mass in each eligible provider world, yet its overall MAE exceeded V3b FULL343. This does not isolate member count because the candidate-fitting and weighting pipelines changed.

Richer disclosure is not sufficient by itself. The I2 development variants were matched in broad search effort but changed loss functions and used an already studied cell. Their results test implemented pipelines, not the optimal value of their disclosed statistics.

Graph structure matters without a simple universal directional rule. A registered serial-versus-parallel masking prediction did not retain one direction across eligible worlds. The finite design does not identify a universal composition-graph error mechanism.

Numerical convergence is insufficient for dependability. Larger graph budget can reduce finite simulation noise without changing bias due to model/search error. No threshold band yet covers the total uncertainty required for a certified real-service admission decision.

11. Threats to validity and reproducibility

Finite physical setting. The core study uses three providers and four logical ASTs on a fixed physical embedding. It does not generalize by itself to arbitrary service graphs, cross-region placement, multi-tenant failures, dynamic routing or real production deployments. The provider generators and fitted process family are related; robustness to substantial model misspecification is not established.

Limited observables. I1 is deliberately compressed and empirical. Region construction was calibrated in frozen simulated worlds; its disclosure noise and dependence on the acquisition policy can affect inverse fits. Quality and power are part of the broader conceptual LCQ/continuum setting but not demonstrated as independent varying stress dimensions here.

I1 acquisition verification. The I1 acquisition procedure was checked against its frozen implementation and provenance records, but the complete acquisition was not independently reproduced from raw simulation events. Moreover, the actual coverage of the constructed regions in the generating provider worlds was not separately validated against the target probability content of the fitted latency-cost models.

Condition selection and missing P2. The Phase-5 registered Step-0 eligibility gate failed for P2 and removed four of 16 intended physical cells. Main 12-cell estimates are **conditional on eligibility**, not estimates of all planned or unseen worlds. The P2-derived retrospective tests use new query regions and are not missing-data imputation.

Method changes and development leakage. V3, V3b and FULL343 changed model search, selection and support after earlier development evidence. The final graph WB was held back behind a hashed global prediction freeze, and archived timestamps/hash links were checked, but this does not restore the stronger status of having all methods fixed before every component of experimental evidence was observed. The same cell was reused while developing Phase-6 I2 variants.

Retrospective original-style M3. The B1400 Top14 and Top1 comparison was specified after prior V3b outcomes were known and reused the same reference. Different provider proposal/fit and local costs prevent causal attribution to ensemble size. The archived audit omits raw native graph ledgers, so it verifies included results rather than bitwise replay.

Budget and correlation. M1/M2/M3 graph budgets differ (100/2700/1400); their numbers of locally simulated trajectories and method fitting time differ as well. The 555 and 6,660 SLA measurements reuse simulation trajectories and query definitions, so they are not independent. Reference bootstrap and per-method numerical error do not simultaneously quantify scenario selection, model-search and physical-world extrapolation uncertainty.

Engineering and software provenance. The modified composition controller is in an external AICon/YAFS repository. The local current dependency revision and source SHA fingerprints were recorded, but exact clean-tree/source-lock provenance at each archived October run was not recovered. Several required smoke checks were completed only after early scientific work, a sequencing deviation documented in the audit. The Phase-5 graph simulator received optional trace-exposure updates after main runs, so current repository HEAD alone does not identify the historical program state.

Numerical evidence tier. Primary saved result summaries and selected hashes have been independently checked against existing pointwise artifacts. This is not complete bitwise replay from every raw YAFS request event. Some ancillary historical pilot operating-point

curves and an M2 paired-bootstrap width remain as reported source claims rather than independently recomputed numerical evidence. No statement above requires inventing their raw observations.

External literature and coauthor consent. The bibliography is inherited from the source manuscript and is **provisional**. References, citation details, current related work and venue overlap policy must be verified before any public release. This draft is internal and does not constitute coauthor approval for arXiv posting.

12. Conclusions and falsifiable next questions

Across Phase I (controlled pilot), Phase II (expanded battery) and Phase III (retrospective divergence tracing), the PRAISE program implemented a discrete-event native composition benchmark, finite public provider SLA disclosures and multiple reconstructed-model integration procedures. It established robust provenance for important frozen outputs, including a hash-frozen prediction/reference separation in the broad Phase-5 battery. The favorable single-world pilot numerical findings were genuine, but the broader evaluation showed **material residual prediction errors** and did not establish general reconstruction sufficiency.

The retrospective M3 comparison shows that near-complete model-bank mass coverage cannot ensure external SLA accuracy. Phase III then established bounded pilot estimator/native-graph parity and documented altered candidate domains and transferred rather than reselected $\lambda = 30$. These are verified protocol differences, not a unique causal explanation. A new holdout with controlled reconstruction ablations is needed to distinguish the causes.

The most actionable scientific questions now concern mechanisms rather than adding model members indiscriminately:

1. **Search recoverability:** When true generating parameters lie inside the search domain and fit I1 markedly better, which proposal distributions, loss landscapes or structural changes allow a public-only learner to find equivalent-quality fits with modest budget? This must be tested with a frozen search protocol and separate end-to-end graph reference.
2. **Information conditional value:** Can a fixed reconstruction objective/architecture be defined so that I1 and I2 disclosures differ *only* in the information made available, permitting an interpretable matched-information comparison?
3. **Graph sensitivity:** Which structural or workload conditions amplify small provider-level errors into end-to-end SLA error, and can the resulting condition be predicted using **publicly observable** quantities rather than privileged white-box outcomes?
4. **Finite-model approximation:** Under what observable diagnostics does Top1 suffice, when do additional hypotheses help and when does weighting mislead? Any rule proposed from a development world must be frozen and tested on other worlds.
5. **Decision calibration:** Can a method produce externally validated error coverage for the hidden graph SLA under clearly limited assumptions? This is a separate problem from making the Monte Carlo standard error small.

These questions are **future hypotheses**, not positive claims established by the existing evaluation. As a scholarly technical record, B1 can support a later focused methodological manuscript by providing precise definitions, method versions, extended experiments and explicit negative outcomes. Publication as a citable preprint remains a subsequent coauthor and venue-policy decision.

Appendix A. Mathematical and numerical interpretation

A.1 Finite-portfolio admission envelopes

REVISIT APPENDIX A.1 WHEN REVIEWING THIS SECTION.
The historical β-based admission-envelope interpretation below is retained provisionally. Decide whether to remove it or describe the finite-model extrema as a range of SLA-satisfaction predictions. Do not treat β as part of the contracted SLA.

For a specified, finite method-generated set $\mathcal{M}_M$ of joint stochastic models, define

$$\sigma^{-}_{G,M}(A,H;\rho) = \min_{\vartheta\in\mathcal{M}_M} \sigma_G(A,H;\rho \mid \vartheta), \qquad \sigma^{+}_{G,M}(A,H;\rho) = \max_{\vartheta\in\mathcal{M}_M} \sigma_G(A,H;\rho \mid \vartheta). \tag{A1}$$

If $\beta \le \sigma^{-}_{G,M}$ then *all enumerated models* accept under their own ideal probabilities; if $\beta > \sigma^{+}_{G,M}$ all reject; otherwise there is disagreement among the finite set. This is a direct statement about the chosen set, **not** a calibrated confidence set, structural-identifiability theorem or SLA truth guarantee. Estimates of the extrema from finite graph simulations add MC error that is absent from this idealized algebra.

A.2 Conditional numerical precision and model spread

The general covariance identity (18) holds for fixed finite weights. M2's paired common-seed bootstrap captures cross-model covariance; M3's independent-member variance bound (14) is conditional on the model bank and frozen weight rule. For a fixed sample allocation and truly independent bounded member outcomes, a weighted Hoeffding inequality gives

$$\Pr\big(|\widehat{p} - p_\alpha| \ge \varepsilon\big) \le 2\exp\left(-\frac{2\varepsilon^2}{\sum_j \alpha_j^2/N_j}\right), \tag{A2}$$

yielding a fixed-sample half-width

$$\varepsilon_\delta = \sqrt{\frac{1}{2}\ln(2/\delta)\sum_j \frac{\alpha_j^2}{N_j}}. \tag{A3}$$

This is included as an *analytical consequence of standard bounded-variable concentration*, **not** a newly tested method or a bound on $|p_\alpha - \sigma_G|$. It is not automatically anytime-valid if sample sizes are chosen by repeatedly observing the outcomes.

The historical noise-adjusted model-spread quantities and white-box correction (19)–(21) are expectation-based diagnostics. Positive-part corrections may be convenient but must not be described as exact interval widths for physical model misspecification.

A.3 Numerical-width scaling

The historical precision appendix extrapolated mean **pointwise interval width** approximately as $\overline{W}(B) = c/\sqrt{B}$, holding support, allocation and the simulation experiment fixed, so that a target width $w^\star$ corresponds formally to $B^\star \approx (c/w^\star)^2$. The pilot anchors were approximately M1 B100/width 0.0375; M2 B2700/width 0.0563 (not independently reproduced here from per-seed paired bootstrap arrays); M3 Top14 B1400/width 0.0493 and B2000/width 0.0425, using **interval clipping to [0,1]** for the reported M3 widths. These are illustrative numerical precision projections, **not evaluated accuracy-versus-budget forecasts**, and they exclude provider reconstruction expense and persistent model bias.

Appendix B. Archived source and audit provenance

B.1 Canonical authority hierarchy

For experimental assertions: frozen experiment contract plus source implementation and selected archive manifests/pointwise tables take precedence over later commentary. The numerical checks verify historical saved outputs, not new simulation. For mathematical definitions, the implementation and mathematical audit take precedence over loosely worded pilot-era prose.

The following documents in `first_science/b1/` are the companion reproducibility record:

- `evidence_registry_v1_0.json`, E01–E12, machine-readable study roles, versions, source paths, numerical values and caveats;
- `METHOD_VERSION_CHRONOLOGY_2026-10-16.md`, Phase1–6 and October P2 stage order;
- `PHASE4_VERIFICATION_REVIEW_2026-10-16.md`, independent pilot and Stage-B pointwise checks;
- `VERIFICATION_REVIEW_V0_3_2026-10-16.md`, Phase5/Phase6/P2 verified saved metrics;
- `TABLES_FIGURES_SOURCE_RECONCILIATION_2026-10-16.md`, 22 original source files, Table2–4 and Fig2–3 checks;
- `METHOD_SUPPORT_VERIFICATION_REVIEW_2026-10-16.md`, 132 V3b selection/allocation checks;
- `SIMULATOR_PROVENANCE_AUDIT_2026-10-16.md` and `RUNTIME_PROVENANCE_VERIFICATION_REVIEW_2026-10-16.md`, code scope, smoke sequencing, 94-pass stage/hash audit;
- `UNCERTAINTY_MATHEMATICS_AUDIT_2026-10-16.md`, exact and approximate interval/variance results;
- `B1_M3_RETROSPECTIVE_EVIDENCE_AUDIT_2026-10-20.md`, retrospective original-style M3: 316 pass, zero fail;
- `retrospective_m3/PHASE_III_DIVERGENCE_CONTROL_REGISTER_2026-10-20.md`, bounded D0–D9 forensic results and limitations;
- `FINAL_AUDIT_SIGNOFF_2026-10-16.md`, conditional drafting authorization and unresolved publication gates;
- `MANUSCRIPT_CLAIM_AUDIT_V0_1.csv`, 68 claims and required dispositions.

Audited integrity milestones

Artifact	Verification
Initial B1 inventory	31 source files and 17 data roots available; 233 manifests inventoried
Phase5/6/P2 saved-results reconciliation	399 passed, zero failed
Phase4 pilot and Stage-B reconciliation	60 passed, zero failed
Phase5 V3b member selection/allocation	132 passed, zero failed
Historical pilot source display archive	22/22 collector hashes and 13/13 checked scientific output hashes match
Phase5 runtime/reference chain	94 passed; 12/12 prediction hashes and 12/12 WB freeze backlinks
Retrospective original-style M3 archive	316 passed; all 12 prediction-manifest backlinks verified
Historical source paper	26-page `v11_45` PDF, preserved and not overwritten

B.2 Version-dependent costs and seeds

Reproducing the Phase-5 executed comparison requires the source at the **archived PRAISE code_commit**, the external modified YAFS/AICon dependency, the public I1 provider cards, V3b selected provider-model and joint-support tables, the frozen graph ASTs, original Step-0

eligibility, named seed banks and archived graph prediction manifests. The final-WB source enumerates **54000–54999** per eligible physical cell. M1/M2 share one method-wide graph seed-index bank in the executed design and M3 uses separate rank-specific blocks. No missing seed or archived experiment is reconstructed in this draft.

For a future reproducibility package, the already assembled B1 tools can extract exact manifest and source hashes without touching frozen data. Current installed source fingerprints are useful but cannot replace a historical environment lock if byte-for-byte results are desired.

Appendix C. Provisional literature references

Bibliographic status: The entries below are transcribed or lightly reformatted from the historical October 2 manuscript. They provide a sourced scholarly starting point, **not a newly completed verification of every author, venue, year or DOI**, and do not claim exhaustive coverage of 2026 research.

1. Bissiri, P. G., Holmes, C. C., and Walker, S. G. "A general framework for updating belief distributions." *Journal of the Royal Statistical Society B* **78**(5), 1103–1130 (2016). DOI: 10.1111/rssb.12158.
2. Bergstra, J., Bardenet, R., Bengio, Y., and Kégl, B. "Algorithms for hyper-parameter optimization." *NeurIPS 24*, 2546–2554 (2011).
3. Blake, M. B., Cummings, D. J., Bansal, A., and Bansal, S. K. "Workflow composition of service level agreements for web services." *Decision Support Systems* **53**(1), 234–244 (2012). DOI: 10.1016/j.dss.2012.01.017.
4. Chen, Y. "Web services composition with incomplete QoS information." *CIT Workshops 2008*, 652–656. DOI: 10.1109/CIT.2008.Workshops.13.
5. De Giacomo, G., De Masellis, R., and Patrizi, F. "Composition of partially observable services exporting their behaviour." *ICAPS 2009*, 90–97. DOI: 10.1609/icaps.v19i1.13360.
6. Geebelen, D., Geebelen, K., Truyen, E., Michiels, S., Suykens, J. A. K., Vandewalle, J., and Joosen, W. "QoS prediction for web service compositions using kernel-based quantile estimation with online adaptation of the constant offset." *Information Sciences* **268**, 397–424 (2014). DOI: 10.1016/j.ins.2013.12.063.
7. Gouriéroux, C., Monfort, A., and Renault, E. "Indirect inference." *Journal of Applied Econometrics* **8**(S1), S85–S118 (1993). DOI: 10.1002/jae.3950080507.
8. Jaeger, M. C., Rojec-Goldmann, G., and Mühl, G. "QoS aggregation for web service composition using workflow patterns." *EDOC 2004*, 149–159. DOI: 10.1109/EDOC.2004.1342512.
9. Lera, I., Guerrero, C., and Juiz, C. "YAFS: A simulator for IoT scenarios in fog computing." *IEEE Access* **7**, 91745–91758 (2019). DOI: 10.1109/ACCESS.2019.2927895.
10. Luo, Z., Wang, D., Yin, Y., Ignatius, J., and Cheng, T. C. E. "Service composition and optimal selection in cloud manufacturing under event-dependent distributional uncertainty of manufacturing capabilities." *European Journal of Operational Research* **325**(2), 281–302 (2025). DOI: 10.1016/j.ejor.2025.03.005.
11. Magrans de Abril, I., Rouska, N.-M., Casamayor Pujol, V., and Dustdar, S. "AICon: Agentic intelligence for the computing continuum." *IEEE Internet Computing* **29**(6), 53–62 (2025). DOI: 10.1109/MIC.2025.3645964.
12. Magrans de Abril, I., Sedlak, B., Casamayor-Pujol, V., and Dustdar, S. "Composable autonomic offerings for decentralized service provisioning in the computing continuum." *ApPLIED 2026*, article 3:1–3:9. DOI: 10.1145/3820355.3820395.
13. Mertens, H., Quatmann, T., and Katoen, J.-P. "Compositional reasoning for probabilistic automata with uncertainty." *arXiv:2603.29550* (2026). DOI: 10.48550/arXiv.2603.29550.

14. Razian, M., Fathian, M., Bahsoon, R., Toosi, A. N., and Buyya, R. “Service composition in dynamic environments: A systematic review and future directions.” *Journal of Systems and Software* **188** (2022). DOI: 10.1016/j.jss.2022.111290.

15. Sedlak, B., Casamayor Pujol, V., Magrans de Abril, I., Donta, P. K., Toosi, A. N., and Dustdar, S. “Service orchestration in the computing continuum: Structural challenges and vision.” *IEEE Internet Computing* **30**(2), 88–98 (2026). DOI: 10.1109/MIC.2026.3662064.

16. Wu, Z., Xiong, N., Huang, Y., Gu, Q., Hu, C., Wu, Z., and Hang, B. “A fast optimization method for reliability and performance of cloud services composition application.” *Journal of Applied Mathematics* **2013**, 1–12. DOI: 10.1155/2013/407267.

17. Xu, D. N., Gössler, G., and Girault, A. “Probabilistic contracts for component-based design.” *ATVA 2010*, LNCS 6252, 325–340. DOI: 10.1007/978-3-642-15643-4_24.

18. Zeng, L., Benatallah, B., Ngu, A. H. H., Dumas, M., Kalagnanam, J., and Chang, H. “QoS-aware middleware for web services composition.” *IEEE Transactions on Software Engineering* **30**(5), 311–327 (2004). DOI: 10.1109/TSE.2004.11.

19. Zheng, H., Yang, J., and Zhao, W. “Probabilistic QoS aggregations for service composition.” *ACM Transactions on the Web* **10**(2), 1–36 (2016). DOI: 10.1145/2876513.

20. Gibbs, A. L., and Su, F. E. “On choosing and bounding probability metrics.” *International Statistical Review* **70**(3), 419–435 (2002). DOI: 10.1111/j.1751-5823.2002.tb00178.x.